

Natural Language Processing

Natural language processing is the field of artificial intelligence concerned with enabling computers to understand, interpret, and generate human language. Early approaches to natural language processing relied on hand-crafted rules and grammars, which struggled to capture the flexibility and ambiguity present in natural languages. The shift toward statistical methods in the 1990s, followed by the deep learning revolution in the 2010s, transformed the field dramatically. Word embeddings, such as those produced by Word2Vec, represent words as dense numerical vectors that capture semantic relationships, allowing words with similar meanings to have similar representations. Two common training approaches for word embeddings are continuous bag of words, which predicts a target word from surrounding context words, and skip-gram, which predicts surrounding context words from a target word. Modern natural language processing systems are built on transformer architectures, which use self-attention mechanisms to weigh the importance of different words in a sentence when producing a representation. Large language models trained on vast amounts of text data have demonstrated impressive abilities in tasks such as translation, summarization, question answering, and text generation. Applications of natural language processing are widespread, including chatbots, sentiment analysis tools that determine whether a piece of text expresses positive or negative opinion, and search engines that better understand user queries. Challenges in the field include handling ambiguity, sarcasm, and cultural context, as well as ensuring that generated text remains factually accurate and free from harmful bias.